ASSIGNMENT 3:

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(Q1):

Write a program to implement (a) pointer to an object (b) this pointer. Practice both '.' (dot operator) and '->' (arrow operator).

CODE:

Untitled-1.cpp > ...

```cpp
#include <iostream>
using namespace std;
class Student {
    int marks;
public:
    void set(int marks) {
        this->marks = marks;
    }
    void show() {
        cout << "Marks = " << marks << endl;
    }
};
int main() {
    Student s;
    Student *p = &s;
    s.set(90);
    s.show();
    p->set(95);
    p->show();
    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\HP\Downloads\dsa> cd "c:\Users\HP\Downloads\dsa\" ;
 }
Marks = 90
Marks = 95
PS C:\Users\HP\Downloads\dsa>
```

(Q2):

Write a program to swap private values of two classes using a friend function.

CODE:

Untitled-1.cpp > main()

```cpp
#include <iostream>
using namespace std;
class B;
class A {
    int x;
public:
    A(int a) { x = a; }
    friend void swap(A&, B&);
    void show() { cout << "A = " << x << endl; }
};
class B {
    int y;
public:
    B(int b) { y = b; }
    friend void swap(A&, B&);
    void show() { cout << "B = " << y << endl; }
};
void swap(A &a, B &b) {
    int temp = a.x;
    a.x = b.y;
    b.y = temp;
}
int main() {
    A a(10);
    B b(20);
    swap(a, b);
    a.show();
    b.show();
    return 0;
}
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
}
A = 20
B = 10
PS C:\Users\HP\Downloads\dsa>
```

(Q3)

Write a program to add data objects of two different classes using friend functions.

CODE:

Untitled-1.cpp > main()

```cpp
#include <iostream>
using namespace std;
class B;
class A {
    int x;
public:
    A(int a) { x = a; }
    friend int add(A, B);
};
class B {
    int y;
public:
    B(int b) { y = b; }
    friend int add(A, B);
};
int add(A a, B b) {
    return a.x + b.y;
}
int main() {
    A a(15);
    B b(25);
    cout << "Sum = " << add(a, b) << endl;
    return 0;
}
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\HP\Downloads\dsa> c:\Users\HP\Downloads\dsa\Untitled-1.cpp
PS C:\Users\HP\Downloads\dsa> cd "c:\Users\HP\Downloads\dsa\" ; if ($?) { g++
Sum = 40
PS C:\Users\HP\Downloads\dsa>
```

(Q4)

Write a program to demonstrate the working of friend class.

CODE:

Untitled-1.cpp > ...

```cpp
#include <iostream>
using namespace std;
class B;
class A {
    int x = 50;
    friend class B;
};
class B {
public:
    void show(A a) {
        cout << "Value of x = " << a.x << endl;
    }
};
int main() {
    A a;
    B b;
    b.show(a);
    return 0;
}
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\HP\Downloads\dsa> c
> cd "c:\Users\HP\Downloads\dsa\" ; if ($?)
Value of x = 50
PS C:\Users\HP\Downloads\dsa>
```

(Q5)

Write a program using Array of Objects to display area of multiple rectangles.

CODE:

Untitled-1.cpp > main()

```cpp
#include <iostream>
using namespace std;
class Rectangle {
    int l, b;
public:
    void input() {
        cin >> l >> b;
    }
    void area() {
        cout << "Area = " << l * b << endl;
    }
};
int main() {
    Rectangle r[3];
    for(int i = 0; i < 3; i++) {
        cout << "Enter length and breadth: ";
        r[i].input();
    }
    for(int i = 0; i < 3; i++) {
        r[i].area();
    }
    return 0;
}
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
Enter length and breadth: 20 5
Enter length and breadth: 2 5
Enter length and breadth: 10 6
Area = 100
Area = 10
Area = 60
PS C:\Users\HP\Downloads\dsa>
```

(Q6)

Write a program to define function ***cube()*** as inline for calculating cube of a number.

CODE:

Untitled-1.cpp > ...

```cpp
#include <iostream>
using namespace std;
inline int cube(int x) {
    return x * x * x;
}
int main() {
    int n;
    cin >> n;
    cout << "Cube = " << cube(n);
    return 0;
}
```

TERMINAL

```
5
Cube = 125
PS C:\Users\HP\Downloads\dsa>
```

(Q7): Write a program to pass an object as an argument and return the object from a function.

a. Use pass-by-value

b. Use pass-by-address

CODE:

(a.)

Untitled-1.cpp

```cpp
#include <iostream>
using namespace std;
class Number {
public:
    int x;
};
Number add(Number n) {
    Number temp;
    temp.x = n.x + 10;
    return temp;
}
int main() {
    Number n1, n2;
    n1.x = 20;
    n2 = add(n1);
    cout << n2.x;
    return 0;
}
```

TERMINAL

```
5
Cube = 125
PS C:\Users\HP\Downloads\dsa> cd "
 ; if ($?) { .\Untitled-1 }
30
PS C:\Users\HP\Downloads\dsa>
```

(b.)

Untitled-1.cpp

```cpp
#include <iostream>
using namespace std;
class Number {
public:
    int x;
};
Number add(Number *n) {
    Number temp;
    temp.x = n->x + 10;
    return temp;
}
int main() {
    Number n1, n2;
    n1.x = 30;
    n2 = add(&n1);
    cout << n2.x;
    return 0;
}
```

TERMINAL

```
PS C:\Users\HP\Downloads\dsa> cd
Untitled-1 }
40
PS C:\Users\HP\Downloads\dsa>
```